

# ETF3231/5231

## Business forecasting

Week 6: Exponential smoothing  
<https://bf.numbat.space/>



# Outline

- 1 ETS models
- 2 Forecasting with ETS models

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**General notation**     E T S : Exponential Smoothing  
                                  ↑    ↑    ↙  
                                 Error Trend Season

```
model(ETS(y ~ error( ) + trend( ) + season( )))
```

**Error:** Additive (A), Multiplicative (M)

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model(ETS(y ~ error( ) + trend( ) + season( )))
```

**Error:** Additive (A), Multiplicative (M)

**Trend:** None (N), Additive (A), Additive damped (Ad).

**Seasonality:** None (N), Additive (A), Multiplicative (M)

## ETS(A,N,N): SES with additive errors

Forecast equation

$$\hat{y}_{t+h|t} = l_t$$

Observation equation

$$y_t = l_{t-1} + \varepsilon_t$$

State equation

$$l_t = l_{t-1} + \alpha \varepsilon_t$$

where  $\varepsilon_t \sim \text{NID}(0, \sigma^2)$ .

- **innovations** or **single source of error** because equations have the same error process,  $\varepsilon_t$ .
- Observation equation: relationship between observations and states.
- State equation(s): evolution of the state(s) through time.

Holt's methods method with additive errors.

Forecast equation

$$\hat{y}_{t+h|t} = \ell_t + hb_t$$

Observation equation

$$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$$

State equations

$$\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t$$

$$b_t = b_{t-1} + \beta\varepsilon_t$$

- Innovation residuals:  $\hat{\varepsilon}_t = y_t - \hat{y}_{t|t-1}$

Holt-Winters additive method with additive errors.

|                      |                                                                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Forecast equation    | $\hat{y}_{t+h t} = \ell_t + hb_t + s_{t+h-m(k+1)}$                                                                                     |
| Observation equation | $y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$                                                                                 |
| State equations      | $\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t$<br>$b_t = b_{t-1} + \beta\varepsilon_t$<br>$s_t = s_{t-m} + \gamma\varepsilon_t$ |

- Innovation residuals:  $\hat{\varepsilon}_t = y_t - \hat{y}_{t|t-1}$
- $k$  is integer part of  $(h - 1)/m$ .

Holt-Winters multiplicative method with multiplicative errors.

|                      |                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forecast equation    | $\hat{y}_{t+h t} = (\ell_t + hb_t)s_{t+h-m(k+1)}$                                                                                                                    |
| Observation equation | $y_t = (\ell_{t-1} + b_{t-1})s_{t-m}(1 + \varepsilon_t)$                                                                                                             |
| State equations      | $\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha\varepsilon_t)$<br>$b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$<br>$s_t = s_{t-m}(1 + \gamma\varepsilon_t)$ |

- Innovation residuals:  $\hat{\varepsilon}_t = (y_t - \hat{y}_{t|t-1})/\hat{y}_{t|t-1}$
- $k$  is integer part of  $(h - 1)/m$ .

# ETS model specification

```
ETS(y ~ error("A") + trend("N") + season("N"))
```

By default, optimal values for  $\alpha$ ,  $\beta$ ,  $\gamma$ , and the states at time 0 are used.

The values for  $\alpha$ ,  $\beta$  and  $\gamma$  can be specified:

```
trend("A", alpha = 0.5, beta = 0.2)
trend("A", alpha_range = c(0.2, 0.8), beta_range = c(0.1, 0.4))
season("M", gamma = 0.04)
season("M", gamma_range = c(0, 0.3))
```

# Exponential smoothing methods

|                 |                   | Seasonal Component  |                     |                     |
|-----------------|-------------------|---------------------|---------------------|---------------------|
|                 |                   | N                   | A                   | M                   |
| Trend Component |                   | (None)              | (Additive)          | (Multiplicative)    |
| N               | (None)            | (N,N)               | (N,A)               | (N,M)               |
| A               | (Additive)        | (A,N)               | (A,A)               | (A,M)               |
| A <sub>d</sub>  | (Additive damped) | (A <sub>d</sub> ,N) | (A <sub>d</sub> ,A) | (A <sub>d</sub> ,M) |

(N,N): Simple exponential smoothing

(A,N): Holt's linear method

(A<sub>d</sub>,N): Additive damped trend method

(A,A): Additive Holt-Winters' method

(A,M): Multiplicative Holt-Winters' method

(A<sub>d</sub>,M): Damped multiplicative Holt-Winters' method

# Exponential smoothing methods

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| A <sub>d</sub>  | (Additive damped) | (A <sub>d</sub> ,N) | (A <sub>d</sub> ,A) | (A <sub>d</sub> ,M) |

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There are also multiplicative trend methods (not recommended).

# ETS models

## Additive Error

|                 |                   | Seasonal Component  |                     |                     |
|-----------------|-------------------|---------------------|---------------------|---------------------|
| Trend Component |                   | N                   | A                   | M                   |
|                 |                   | (None)              | (Additive)          | (Multiplicative)    |
| N               | (None)            | A,N,N               | A,N,A               | A,N,M               |
| A               | (Additive)        | A,A,N               | A,A,A               | A,A,M               |
| A <sub>d</sub>  | (Additive damped) | A,A <sub>d</sub> ,N | A,A <sub>d</sub> ,A | A,A <sub>d</sub> ,M |

## Multiplicative Error

|                 |                   | Seasonal Component  |                     |                     |
|-----------------|-------------------|---------------------|---------------------|---------------------|
| Trend Component |                   | N                   | A                   | M                   |
|                 |                   | (None)              | (Additive)          | (Multiplicative)    |
| N               | (None)            | M,N,N               | M,N,A               | M,N,M               |
| A               | (Additive)        | M,A,N               | M,A,A               | M,A,M               |
| A <sub>d</sub>  | (Additive damped) | M,A <sub>d</sub> ,N | M,A <sub>d</sub> ,A | M,A <sub>d</sub> ,M |

# Exponential smoothing models

## Additive Error

|                 |                   | Seasonal Component  |                     |                              |
|-----------------|-------------------|---------------------|---------------------|------------------------------|
| Trend Component |                   | N<br>(None)         | A<br>(Additive)     | M<br>(Multiplicative)        |
| N               | (None)            | A,N,N               | A,N,A               | <del>A,N,M</del>             |
| A               | (Additive)        | A,A,N               | A,A,A               | <del>A,A,M</del>             |
| A <sub>d</sub>  | (Additive damped) | A,A <sub>d</sub> ,N | A,A <sub>d</sub> ,A | <del>A,A<sub>d</sub>,M</del> |

## Multiplicative Error

|                 |                   | Seasonal Component  |                     |                       |
|-----------------|-------------------|---------------------|---------------------|-----------------------|
| Trend Component |                   | N<br>(None)         | A<br>(Additive)     | M<br>(Multiplicative) |
| N               | (None)            | M,N,N               | M,N,A               | M,N,M                 |
| A               | (Additive)        | M,A,N               | M,A,A               | M,A,M                 |
| A <sub>d</sub>  | (Additive damped) | M,A <sub>d</sub> ,N | M,A <sub>d</sub> ,A | M,A <sub>d</sub> ,M   |

# Additive error models

| Trend          | Seasonal                                                                                                                                                       |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | N                                                                                                                                                              | A                                                                                                                                                                                                                  | M                                                                                                                                                                                                                                                                    |
| N              | $y_t = \ell_{t-1} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \alpha \varepsilon_t$                                                                             | $y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \alpha \varepsilon_t$<br>$s_t = s_{t-m} + \gamma \varepsilon_t$                                                                             | $y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$<br>$s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$                                                                                                          |
| A              | $y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$<br>$b_t = b_{t-1} + \beta \varepsilon_t$                | $y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$<br>$b_t = b_{t-1} + \beta \varepsilon_t$<br>$s_t = s_{t-m} + \gamma \varepsilon_t$                | $y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t / s_{t-m}$<br>$b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$<br>$s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$                     |
| A <sub>d</sub> | $y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$<br>$b_t = \phi b_{t-1} + \beta \varepsilon_t$ | $y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$<br>$b_t = \phi b_{t-1} + \beta \varepsilon_t$<br>$s_t = s_{t-m} + \gamma \varepsilon_t$ | $y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$<br>$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$<br>$b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$<br>$s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$ |

# Multiplicative error models

| Trend          | Seasonal                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | N                                                                                                                                                                                                 | A                                                                                                                                                                                                                                                                                                                                   | M                                                                                                                                                                                                                                                    |
| N              | $y_t = \ell_{t-1}(1 + \varepsilon_t)$<br>$\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t)$                                                                                                           | $y_t = (\ell_{t-1} + s_{t-m})(1 + \varepsilon_t)$<br>$\ell_t = \ell_{t-1} + \alpha(\ell_{t-1} + s_{t-m})\varepsilon_t$<br>$s_t = s_{t-m} + \gamma(\ell_{t-1} + s_{t-m})\varepsilon_t$                                                                                                                                               | $y_t = \ell_{t-1}s_{t-m}(1 + \varepsilon_t)$<br>$\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t)$<br>$s_t = s_{t-m}(1 + \gamma\varepsilon_t)$                                                                                                           |
| A              | $y_t = (\ell_{t-1} + b_{t-1})(1 + \varepsilon_t)$<br>$\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha\varepsilon_t)$<br>$b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$                     | $y_t = (\ell_{t-1} + b_{t-1} + s_{t-m})(1 + \varepsilon_t)$<br>$\ell_t = \ell_{t-1} + b_{t-1} + \alpha(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$<br>$b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$<br>$s_t = s_{t-m} + \gamma(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$                               | $y_t = (\ell_{t-1} + b_{t-1})s_{t-m}(1 + \varepsilon_t)$<br>$\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha\varepsilon_t)$<br>$b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$<br>$s_t = s_{t-m}(1 + \gamma\varepsilon_t)$                     |
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# Model selection

## Akaike's Information Criterion

$$\text{AIC} = -2 \log(L) + 2k$$

where  $L$  is the likelihood and  $k$  is the number of parameters initial states estimated in the model.

## Corrected AIC

$$\text{AIC}_c = \text{AIC} + \frac{2k(k+1)}{T-k-1}$$

which is the AIC corrected (for small sample bias).

## Bayesian (Schwartz) Information Criterion

$$\text{BIC} = \text{AIC} + k[\log(T) - 2] = -2 \log(L) + \ln(T)k$$

Minimizing the AIC assuming Gaussian residuals is asymptotically equivalent to minimizing one-step time series cross validation MSE.

## From Hyndman et al. (IJF, 2002):

- Apply each model that is appropriate to the data. Optimize parameters and initial values using MLE (or some other criterion).
- Select best model using AICc:
- Produce forecasts using best model.
- Obtain forecast intervals using underlying state space model.

Method performed very well in M3 competition.

# Outline

- 1 ETS models
- 2 Forecasting with ETS models

# Forecasting with ETS models

**Traditional point forecasts:** iterate the equations for  $t = T + 1, T + 2, \dots, T + h$ .

# Forecasting with ETS models

**Traditional point forecasts:** iterate the equations for  $t = T + 1, T + 2, \dots, T + h$ .

- Not the same as  $E(y_{t+h}|\mathbf{x}_t)$  unless seasonality is additive.
- fable uses  $E(y_{t+h}|\mathbf{x}_t)$ .
- Point forecasts for  $ETS(A,*,*)$  are identical to  $ETS(M,*,*)$  if the parameters are the same.

## Example: ETS(A,A,N)

$$y_{T+1} = \ell_T + b_T + \varepsilon_{T+1}$$

$$\hat{y}_{T+1|T} = \ell_T + b_T$$

$$y_{T+2} = \ell_{T+1} + b_{T+1} + \varepsilon_{T+2}$$

$$= (\ell_T + b_T + \alpha\varepsilon_{T+1}) + (b_T + \beta\varepsilon_{T+1}) + \varepsilon_{T+2}$$

$$\hat{y}_{T+2|T} = \ell_T + 2b_T$$

etc.

## Example: ETS(M,A,N)

$$y_{T+1} = (\ell_T + b_T)(1 + \varepsilon_{T+1})$$

$$\hat{y}_{T+1|T} = \ell_T + b_T.$$

$$y_{T+2} = (\ell_{T+1} + b_{T+1})(1 + \varepsilon_{T+2})$$

$$= \left\{ (\ell_T + b_T)(1 + \alpha\varepsilon_{T+1}) + \left[ b_T + \beta(\ell_T + b_T)\varepsilon_{T+1} \right] \right\} (1 + \varepsilon_{T+2})$$

$$\hat{y}_{T+2|T} = \ell_T + 2b_T$$

etc.

**Prediction intervals:** can only be generated using the models.

- The prediction intervals will differ between models with additive and multiplicative errors.
- Exact formulae for some models.
- More general to simulate future sample paths, conditional on the last estimate of the states, and to obtain prediction intervals from the percentiles of these simulated future paths.

# Prediction intervals

PI for most ETS models:  $\hat{y}_{T+h|T} \pm c\sigma_h$ , where  $c$  depends on coverage probability and  $\sigma_h$  is forecast standard deviation.

$$(A,N,N) \quad \sigma_h = \sigma^2 \left[ 1 + \alpha^2(h-1) \right]$$

$$(A,A,N) \quad \sigma_h = \sigma^2 \left[ 1 + (h-1) \left\{ \alpha^2 + \alpha\beta h + \frac{1}{6}\beta^2 h(2h-1) \right\} \right]$$

$$(A,A_d,N) \quad \sigma_h = \sigma^2 \left[ 1 + \alpha^2(h-1) + \frac{\beta\phi h}{(1-\phi)^2} \{2\alpha(1-\phi) + \beta\phi\} - \frac{\beta\phi(1-\phi^h)}{(1-\phi)^2(1-\phi^2)} \{2\alpha(1-\phi^2) + \beta\phi(1+2\phi-\phi^h)\} \right]$$

$$(A,N,A) \quad \sigma_h = \sigma^2 \left[ 1 + \alpha^2(h-1) + \gamma k(2\alpha + \gamma) \right]$$

$$(A,A,A) \quad \sigma_h = \sigma^2 \left[ 1 + (h-1) \left\{ \alpha^2 + \alpha\beta h + \frac{1}{6}\beta^2 h(2h-1) \right\} + \gamma k \{2\alpha + \gamma + \beta m(k+1)\} \right]$$

$$(A,A_d,A) \quad \sigma_h = \sigma^2 \left[ 1 + \alpha^2(h-1) + \frac{\beta\phi h}{(1-\phi)^2} \{2\alpha(1-\phi) + \beta\phi\} - \frac{\beta\phi(1-\phi^h)}{(1-\phi)^2(1-\phi^2)} \{2\alpha(1-\phi^2) + \beta\phi(1+2\phi-\phi^h)\} \right. \\ \left. + \gamma k(2\alpha + \gamma) + \frac{2\beta\gamma\phi}{(1-\phi)(1-\phi^m)} \{k(1-\phi^m) - \phi^m(1-\phi^{mk})\} \right]$$